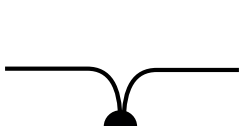
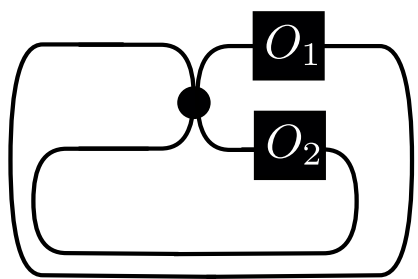
(a)



$= \frac{1}{2}|00\rangle\langle 00| + \frac{1}{2}|11\rangle\langle 11|$

(b)



$= \mathrm{Tr}\left(\rho'_{\mathsf{GHZ}} \cdot O_1 \otimes O_2\right)$